CYBERSENTINEL

PROBLEM STATEMENT

Many organizations struggle with rapidly increasing cyber attacks such as DDoS, brute force, and unauthorized access. Traditional security systems rely heavily on manual monitoring and predefined rules, which are often slow and ineffective against modern, dynamic threats. There is a need for an intelligent, proactive threat hunting system that can continuously analyze network activity, identify abnormal behavior, and suggest responses without constant human intervention.

PROJECT OVERVIEW

CyberSentinel is an AI-powered autonomous threat hunting system that combines real-time network monitoring, automated threat detection, and intelligent response recommendation into a single integrated platform. The system continuously processes network log data, analyzing parameters like packet counts and login attempts to classify network activity. Unlike traditional security tools, CyberSentinel utilizes Agentic AI to independently suggest mitigation strategies, improving overall network resilience and security posture.

SOLUTION OFFERED

CyberSentinel analyzes network traffic patterns to detect suspicious behavior and automatically classifies threats into LOW, MEDIUM, or HIGH risk levels. It identifies specific attack types, such as DDoS or brute force, and uses Agentic AI to recommend immediate actions like blocking suspicious IP addresses. The system features a professional SOC-style dashboard with real-time visualizations and a live attack feed simulation. By providing explainable AI for every classification, the system ensures transparent and efficient security operations management.

END USERS

- **SOC Teams:** Automated threat monitoring.
- **Network Admins:** Simplified traffic analysis.
- **SMEs:** Cost-effective automated defense.
- **Students/Researchers:** Learning anomaly detection.

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TECHNOLOGY USED

Frontend: Streamlit (SOC-style Dashboard), Plotly (Interactive Visualizations)
Backend: Python
Data Processing: Pandas, NumPy
Machine Learning/AI: Agentic AI Framework, Scikit-learn
Algorithms: Random Forest (Classification), Anomaly Detection, Logic-based Response Generation